Simulation Of The Flow Over NACA0012 Airfoil

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Abstract

The Large Eddy Simulation code in OpenFOAM with the Spalart Allmaras Detached Eddy Simulation method is used to simulate the airflow over the NACA0012 airfoil. To the best of the author's knowledge, there are no tutorials available on the Spalart Allmaras DES, neither in the OpenFOAM tutorial folders nor on the internet. This report with the case files in the project directory serve as a guiding tutorial for applying this type of turbulent models.

List of Symbols

$\kappa, C_{b1}, C_{b2}, C_{w1}, \sigma, C_{w2}, C_{w3}, C_{v1}, C_s$	Model constants
ν	Kinematic viscosity $[m^2/s]$
ν_t	Turbulent viscosity $[m^2/s]$
ρ	Density of the fluid $[kg/m^3]$
$\tilde{\nu}$	Eddy viscosity $[m^2/s]$
f_{v1}, f_{v2}, f_w	Model auxiliary functions
$g, r, \tilde{S}$	
p	Kinematic pressure $[m^2/s^2]$
S	Vorticity magnitude
u	Velocity vector $[m/s]$
u_x	Velocity in the x direction $[m/s]$
u_y	Velocity in the y direction $[m/s]$

1. Introduction

Spalart Allmaras (SA) turbulent model is essentially designed for incompressible external flows. The Spalart Allmaras Detached Eddy Simulation (SA-DES) was propsed as an enhancement. In this model, regions near a boundary wall are treated as a Reynolds Averaged Navier Stokes (RANS) problem, while the rest of the flow is solved using a Large Eddy Simulation (LES) approach. To the best of the author's knowledge, there are no tutorials available on the internet that demonstrate the implementation of the model. Hence in this report, the SA-DES model is applied on the flow around a NACA 0012 airfoil.

2. Theory

In this study, the open source software OpenFOAM is used. Several assumptions are made to simulate the flow around a NACA 0012 airfoil. The flow is taken to be turbulent, transient, viscous, two-dimensional, and incompressible. Hence, computations were performed using the transient solver for incompressible flows available in OpenFOAM (PISOFoam). It uses the PISO algorithm for pressure–velocity decoupling.

2.1. Governing Equations

The aforementioned assumptions are translated into the following governing equations:
1. Incompressibility, given by the continuity equation:

$$\nabla . u = 0 \tag{1}$$

2.The transient momentum equation, given by:

$$\frac{\partial u}{\partial t} + \nabla . (u \otimes u) = -\nabla p \tag{2}$$

Equations (1) and (2) are accompanied with the initial and boundary conditions given in table (1)

Initial & Boundary Conditions
$u_x = 25.75$ m/s
$u_y = 3.62$ m/s
$p = 0$ Pa

Table 1: Initial and boundary conditions for equations (1) and (2).

2.2. Turbulence Modelling

The PISOFoam solver -which uses LES- is used with the SA-DES turbulence model. The original SA-DES model was introduced by Spalart et al. The model replaces the minimum wall distance d in the SA–RANS model with a modified length scale $\tilde{d}$ given by:

$$\tilde{d} = min(d, C_{DES}\Delta)$$

where:
$\Delta = max(\Delta x, \Delta y, \Delta z)$
$\Delta x, \Delta y, \Delta z$ are the grid sizes.
C_{DES} is a constant.

This is how the model works: in the boundary layer, the normal wall distance d is much smaller than the wall parallel grid spacing. The modified normal wall distance consequently acts as a length scale, $\tilde{d} = d$, and the SA–RANS model is used within the boundary layer. On the other hand, outside the boundary layer, the normal wall distance is larger than the grid spacing, so the modified length scale becomes $\tilde{d} = C_{DES}$. Since it depends on the grid size, this length scale enables the DES model to work like LES in separated flow regions, which are typically located away from the walls. To complete the set of the PDE (equations (1) and (2)), an equation for the modified turbulence viscosity ($\tilde{\nu}$)) is added. This equation is given by:

$$\frac{\partial \tilde{\nu}}{\partial t} = \nabla.(u\tilde{\nu}) + C_{b1}\tilde{S}\tilde{\nu} + \frac{C_{b2}}{\sigma}\nabla\tilde{\nu}.\nabla\tilde{\nu} + \\ + \frac{1}{\sigma}\nabla.((\nu + \tilde{\nu})\nabla\tilde{\nu}) - C_{w1}f_w\left(\frac{\tilde{\nu}}{\tilde{d}}\right)^2 \tag{3}$$

The turbulence viscosity is obtained using:

$$\nu_t = \bar{\nu} f_{v1} \tag{4}$$

The following closure functions are used to complete the definition of the PDE system of equations ((1), (2) and (3)):

$$f_{v1} = \frac{\chi^3}{\chi^3 + {C_{v1}}^3}, \qquad f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}}$$

$$f_w = g\left[\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6}\right]^{\frac{1}{6}}, \qquad g = r + C_{w2}(r^6 - r)$$

$$r = \frac{\tilde{\nu}}{\tilde{S}k^2d^2}, \qquad \tilde{S} = S + \frac{\tilde{\nu}}{\tilde{S}k^2d^2}f_{v2}$$

Where:
$\chi = \tilde{\nu}/\nu$
$S = \sqrt{2\Omega : \Omega}$
$\Omega = \frac{1}{2}(\nabla u - \nabla u^T)$

The closure parameters are summarized in table (2).

Coefficient	Value
σ	2/3
C_{b1}	0.1355
C_{b2}	0.622
C_{w1}	$\frac{C_{b1}}{k^2} + \frac{1+C_{b2}}{\sigma_{\nu_t}}$
C_{w2}	0.3
C_{w3}	2
C_{v1}	7.1
C_s	0.3
κ	0.41

Table 2: SA-DES closure parameters.

To solve equation (3), it should be accompanied with the initial values for ν and $\tilde{\nu}$, given by:

$$\nu = \tilde{\nu} = 0.14$$

3. Results and Discussion

The grid used in this study has a 10,720 total number of cells. The grid is shown in figures (1) and (2).

As seen from the velocity magnitude and since the angle of attack is zero, the stagnation point is at the airfoil leading edge. Figure (3)

As expected and as the Bernoulli equation dictates, pressure values are low where the velocity values are high (upper camber). On the other hand, pressure values are high where the velocity values are low (leading and trailing edges). see figure (4)

Instantaneous values of $\tilde{\nu}$ are given in figure (5). It can be seen that values are at their lowest at the stagnation point.

Streamlines are shown in figure (6). It can be seen that at angle of attack zero, air flows smoothly over the airfoil i.e. no flow separation occurs, and hence no recirculation is noticed.

4. References

1.https://www.openfoam.com/documentation/guides/latest/doc/verification-validation-naca0012-airfoil-2d.html

2.https://www.openfoam.com/documentation/guides/latest/doc/guide-turbulence-ras-spalart-allmaras.html

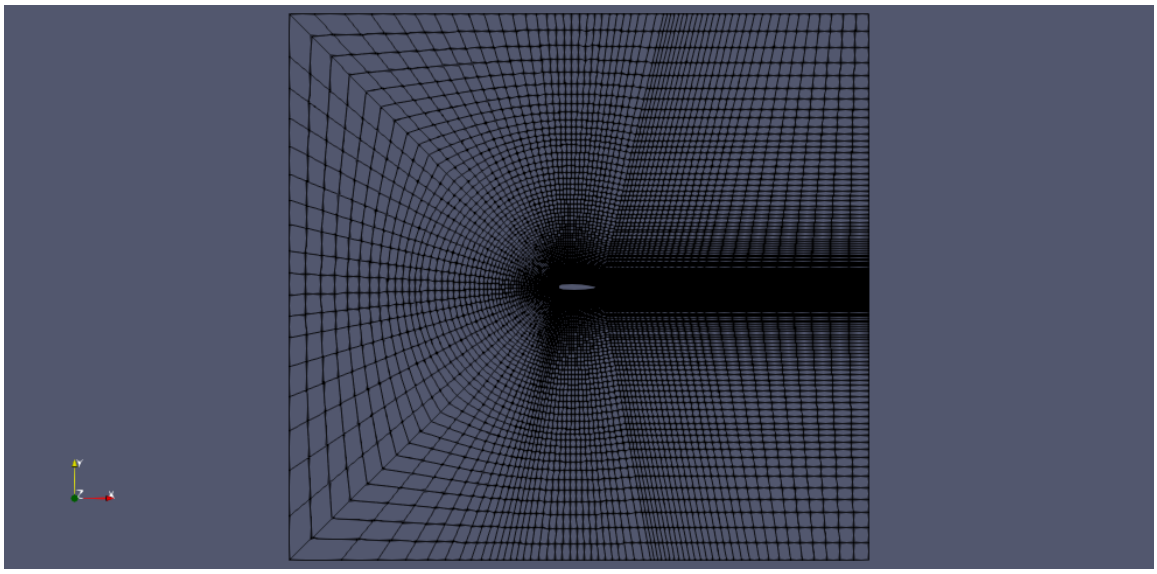

Figure 1: Far field computational grid.

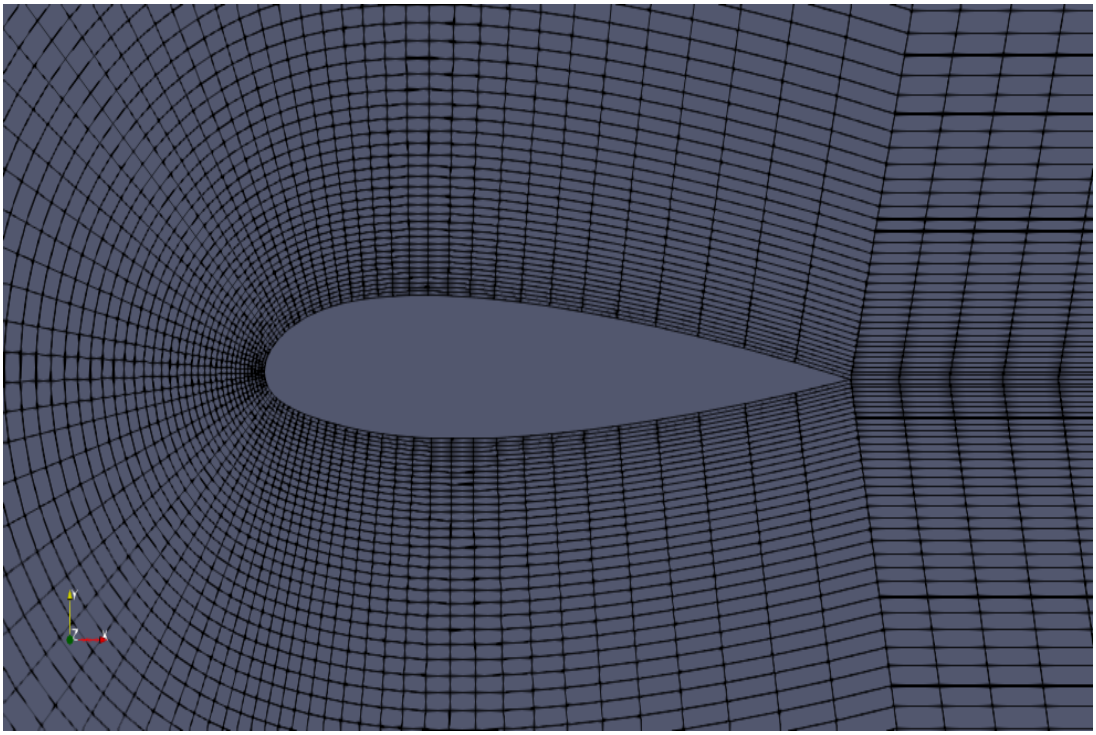

Figure 2: Computational grid near the airfoil.

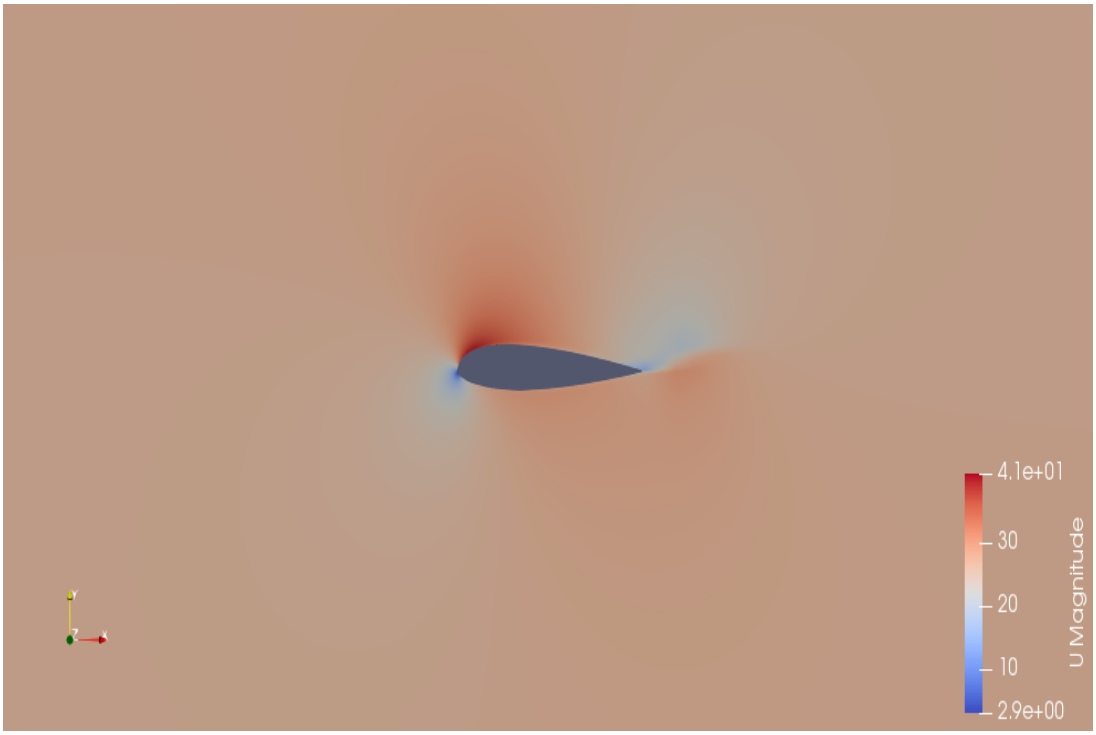

Figure 3: Velocity instantaneous values.

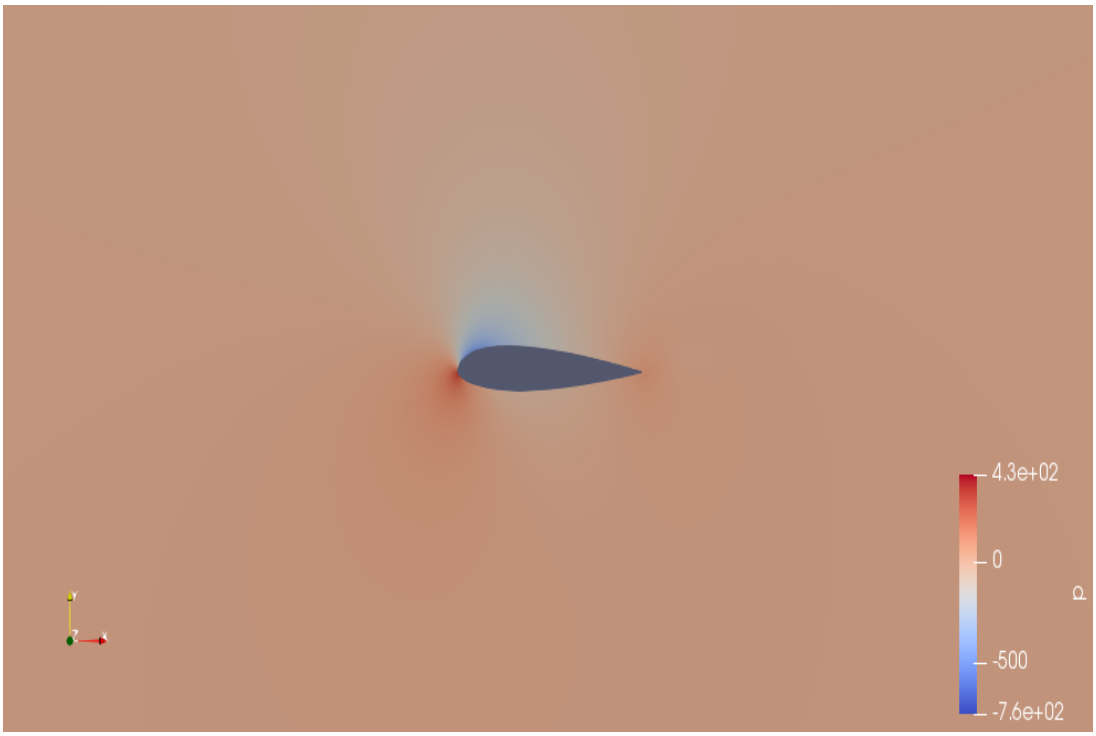

Figure 4: Pressure instantaneous values.

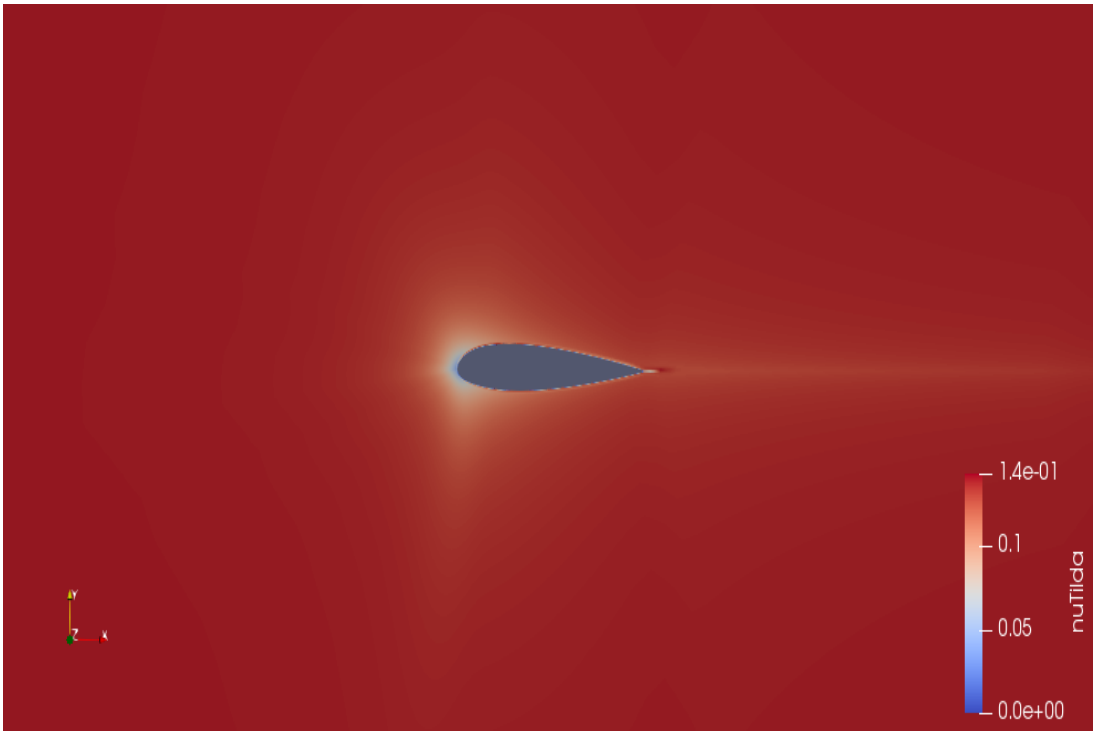

Figure 5: Modified turbulent viscosity $\tilde{\nu}$ instantaneous values.

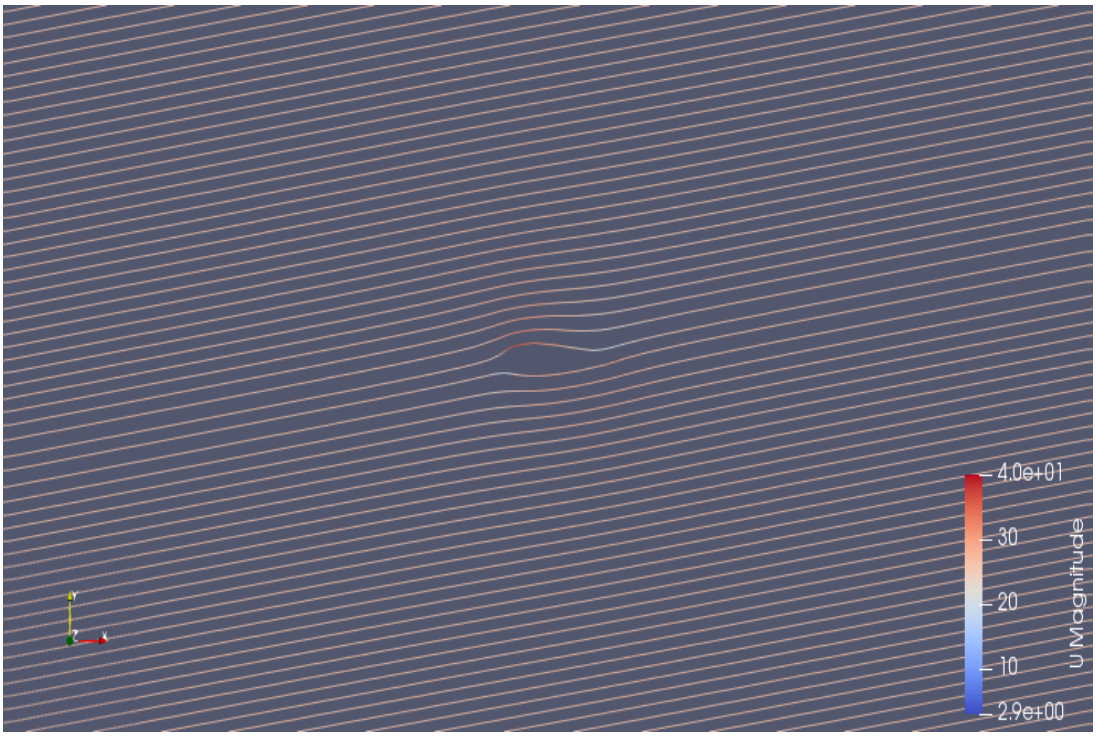

Figure 6: Streamlines of the instantaneous velocity.

3. Detached eddy simulations of the flow over a cylinder at Re= 3900 using OpenFOAM, D'Alessandro et.al, Computers Fluids, volume 136, pages 152-169, 2016, Elsevier.

4.https://resources.system-analysis.cadence.com/blog/msa2021-what-is-the-spalart-allmaras-turbulence-model